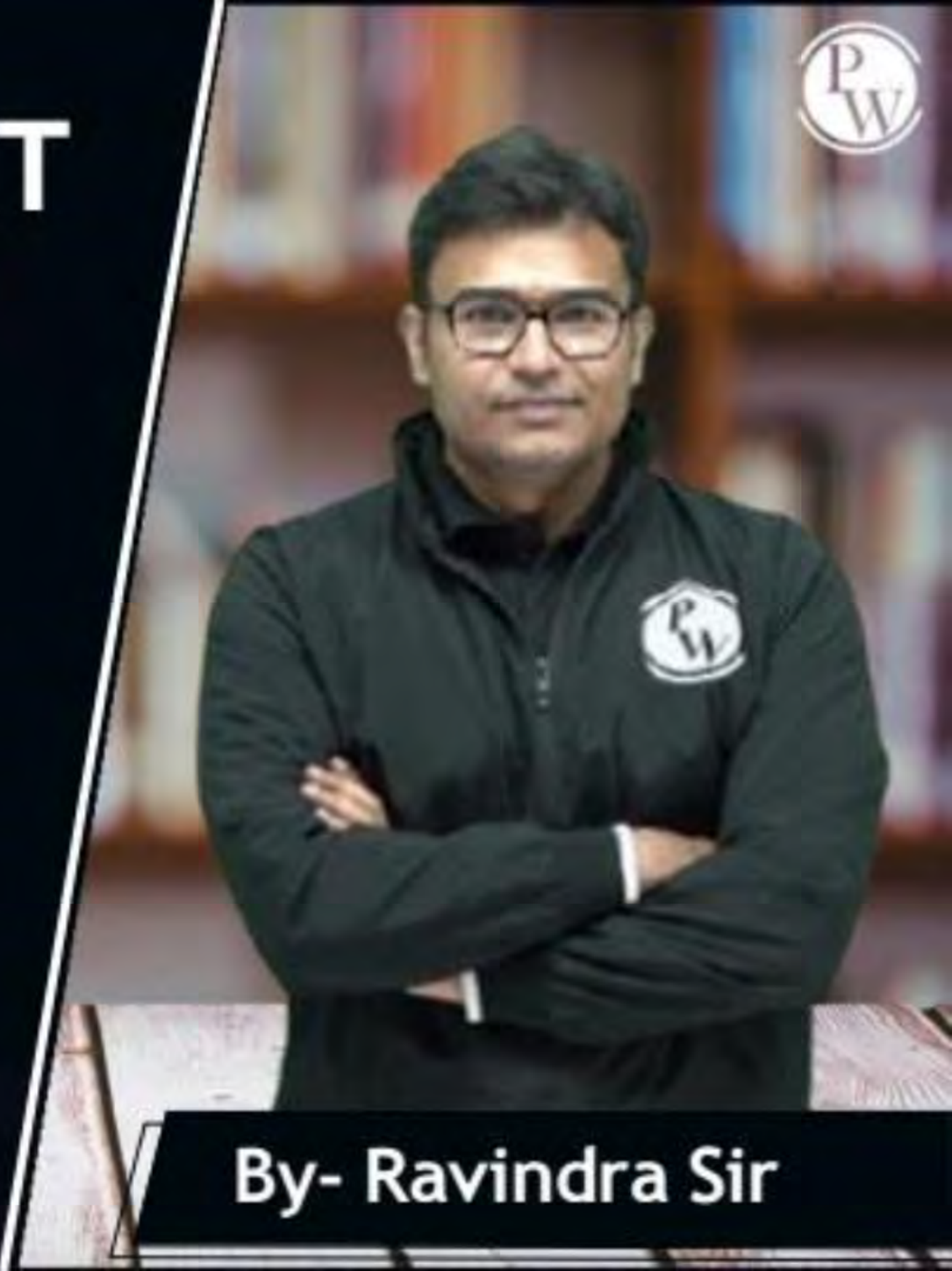
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 03



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

layers

Topics to be Covered



- Topic
- Topic
- Topic

Switching

Inspiring Stories : Hariman Sharma



Background: Farmer from Himachal, no training, just love for farming.

Education: Only basic school

Achievements: Grew a new apple that can survive in hot places.

Impact: Helped farmers in warm regions earn more money.

Inspiring Stories : Matmur Jamoh



Background: Born in 1880s in Arunachal Pradesh.

Education: No formal schooling, grew up in a tribal community.

Achievements: In 1911, led his people in the Kekar Monying resistance, attacking British officers who tried to control their land.

Impact: Became a symbol of courage, remembered for defending his land against the British.

Inspiring Stories : S. V. Setty

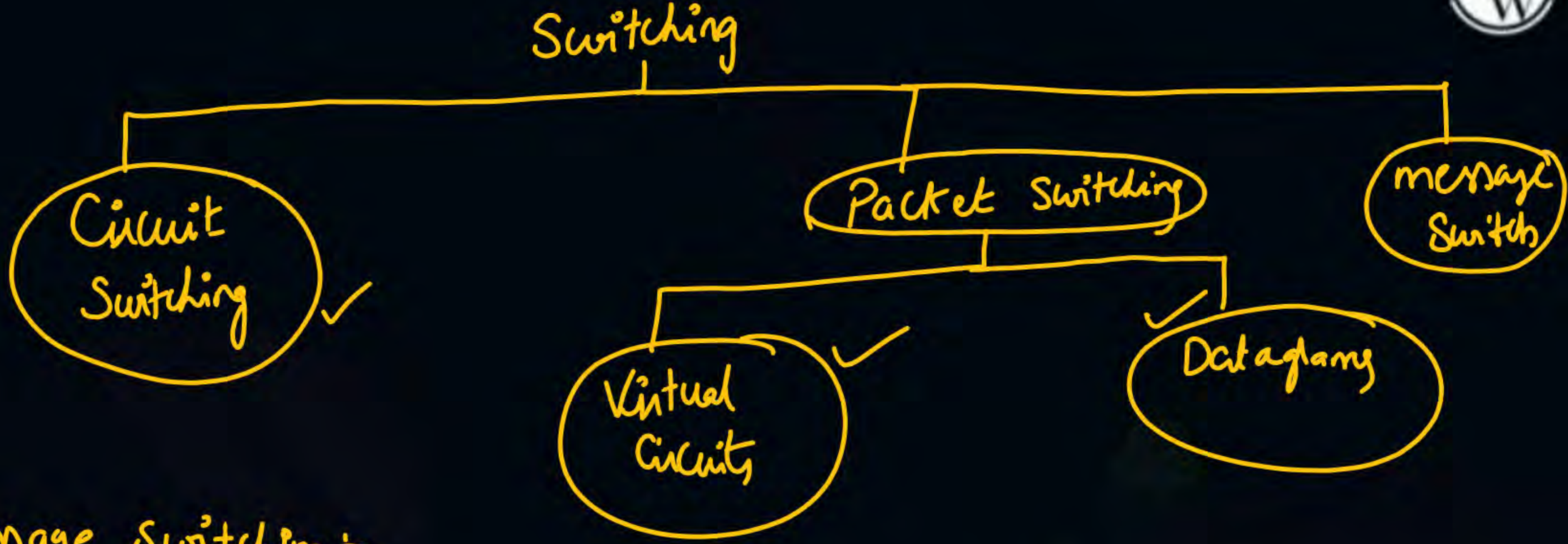


Background: Born in Mysore, simple family.

Education: Studied engineering in India and UK.

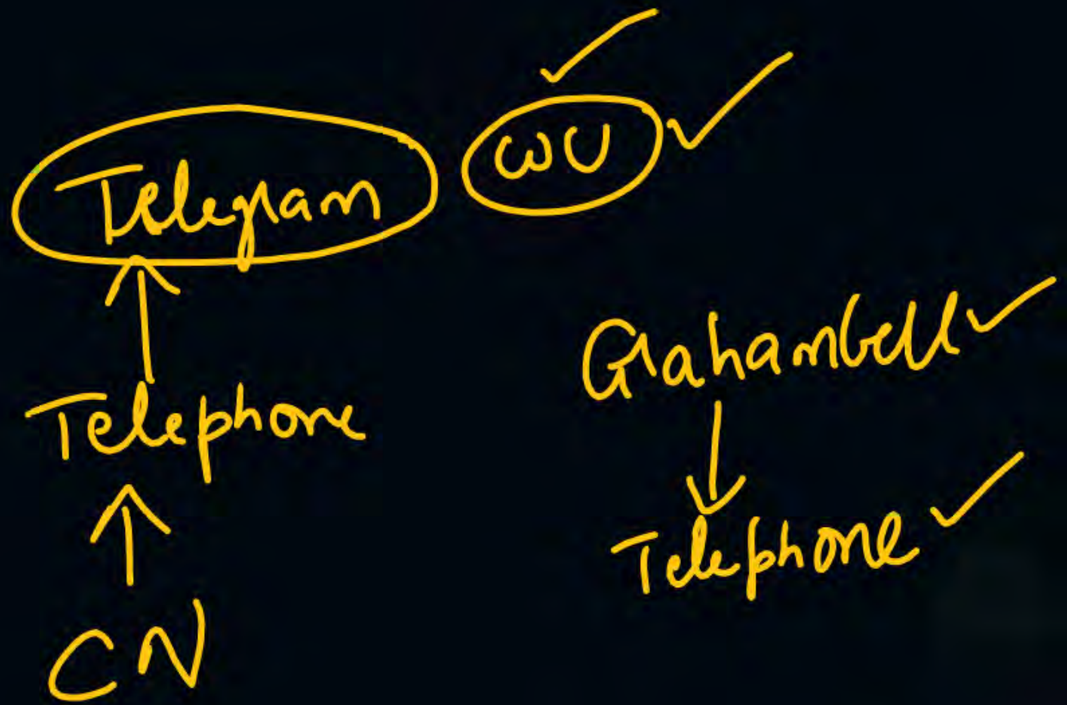
Achievements: Helped design a famous war plane (Avro 504) and started Bengaluru's first engineering college.

Impact: Opened doors for Indian aviation and trained future engineers.

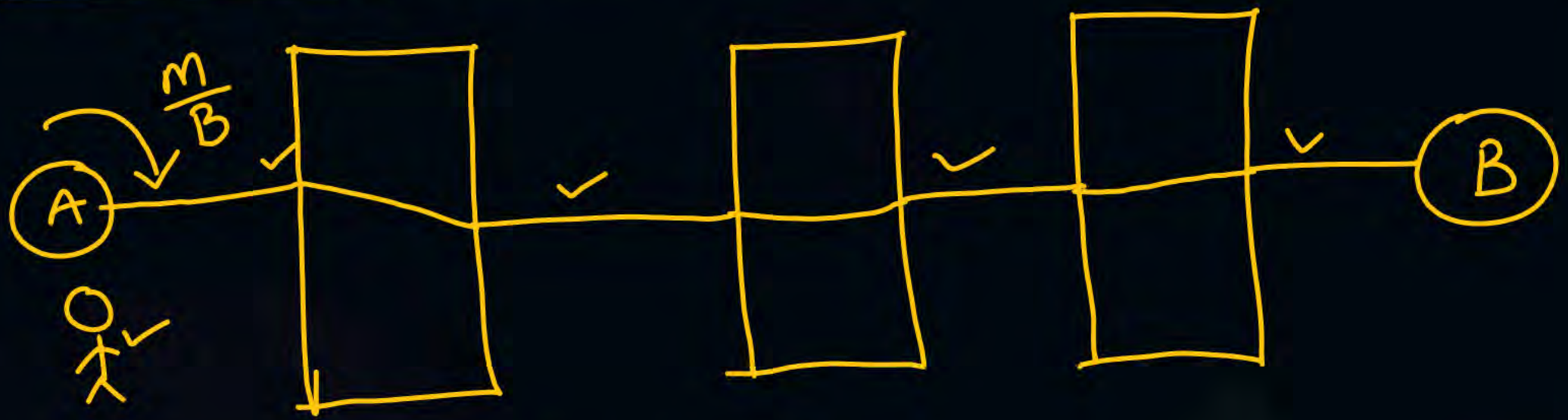


Message Switching :-

message is not divided into packets.


Telegram
WU
Telephone
CN
Grahambell
Telephone

Circuit Switching :- (PL) → obsolete

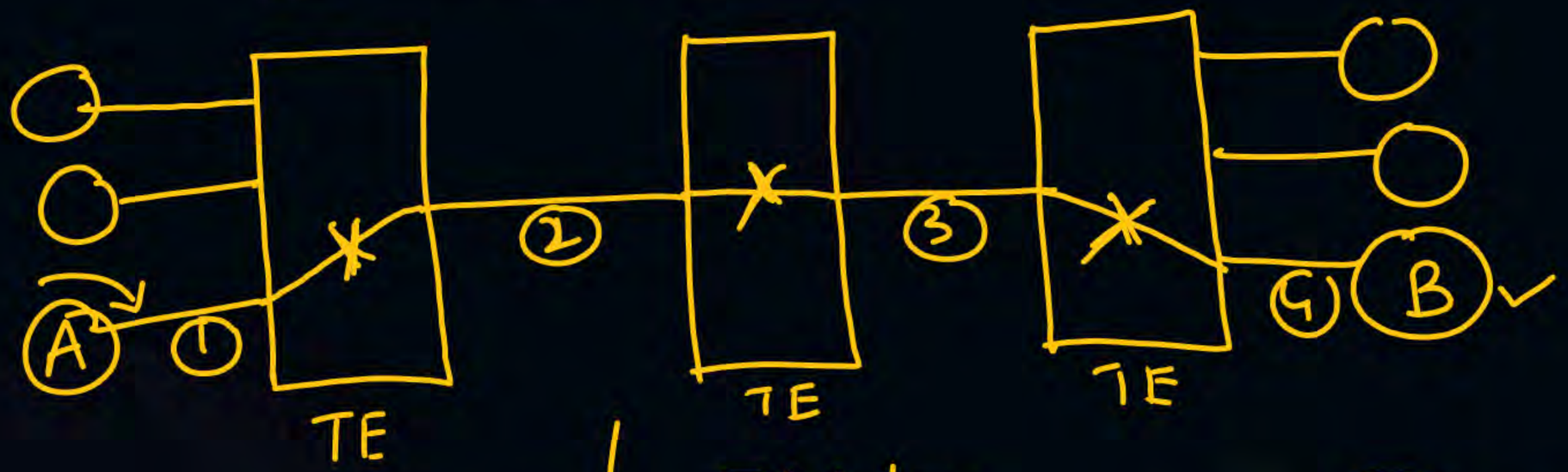


M - length of the data
X → No of hops
B → B/W
d → length of a hop
v → Velocity of a signal

Total time :-

$$\text{Setup time} + \frac{M}{B} + X\frac{d}{v} + \text{Teardown time}$$

Circuit Switching



M - Message
B - B/w
X - No of hops
d → distance of each hop
V → velocity

Total time
= Setup time + $\frac{M}{B}$ + $X\frac{d}{V}$ + Teardown time

PL → Obsolete

Packet Switching (Store and Forward)



M - message
B - BW
X - no of hops
d - distance
V - velocity

Total time = $\frac{X m}{B} + \frac{X d}{V}$

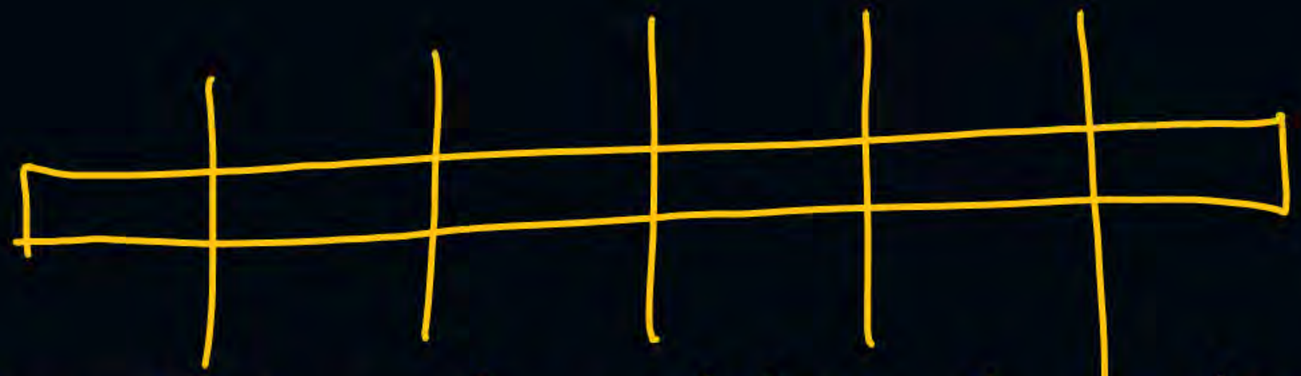
CS ✓	PS ✓
ST + TD	$(X-1)\frac{m}{B}$ ✓



M → Bursty

PS X

CS ✓



Packetization is a process of dividing big message into small parts. This will have adv of pipeling



Data = 1000 B

BW = 1 mBPS

= 10^6 BPS

Header = 100 B

Assume → $T_p = 0$

$L = 1000 + 100$

$= 1100$ B

$T_t = L/B = \frac{1100}{10^6} = 1.1$ ms

Total time = $3 \times 1.1 =$ 3.3 msec



Packets = 5 packets ✓

Data = $\frac{1000}{5}$ = 200B

BW = 1mBps

Header = 100B

Packet Size = 200B + 100B = 300B

$T_t = \frac{300}{10^6} = 0.3$ mS

I-packet = 3 × 0.3 mS = 0.9 mS

4 packets = 4 × 0.3 mS = 1.2 mS

Total time = 0.9 + 1.2 mS = 2.1 mS ✓



Packets = 10 ✓

Data = $\frac{1000}{10}$ = 100B

header = 100 B

BW = 1 MBPS

L = 100 + 100 = 200B

$T_t = L/B = \frac{200}{10^6} = 0.2$ ms

I → 3 × 0.2 ms = 0.6 ms

9 packets − 9 × 0.2 ms = 1.8 ms

TT = 0.6 ms + 1.8 ms

= 2.4 ms ✓



Packets = 20 ✓

Data = $\frac{1000}{20}$ = 50B

Header = 100B ✓

L = 50 + 100 = 150B

$T_t = L/B = 150/10^6 = 0.15$ ms

I packet = 3 × 0.15 ms
= 0.45 ms

19 packet = 19 × 0.15 ms
= 1.85 ms

Total time = 0.45 + 1.85 ms
= 3.3 ms

Virtual circuits:-



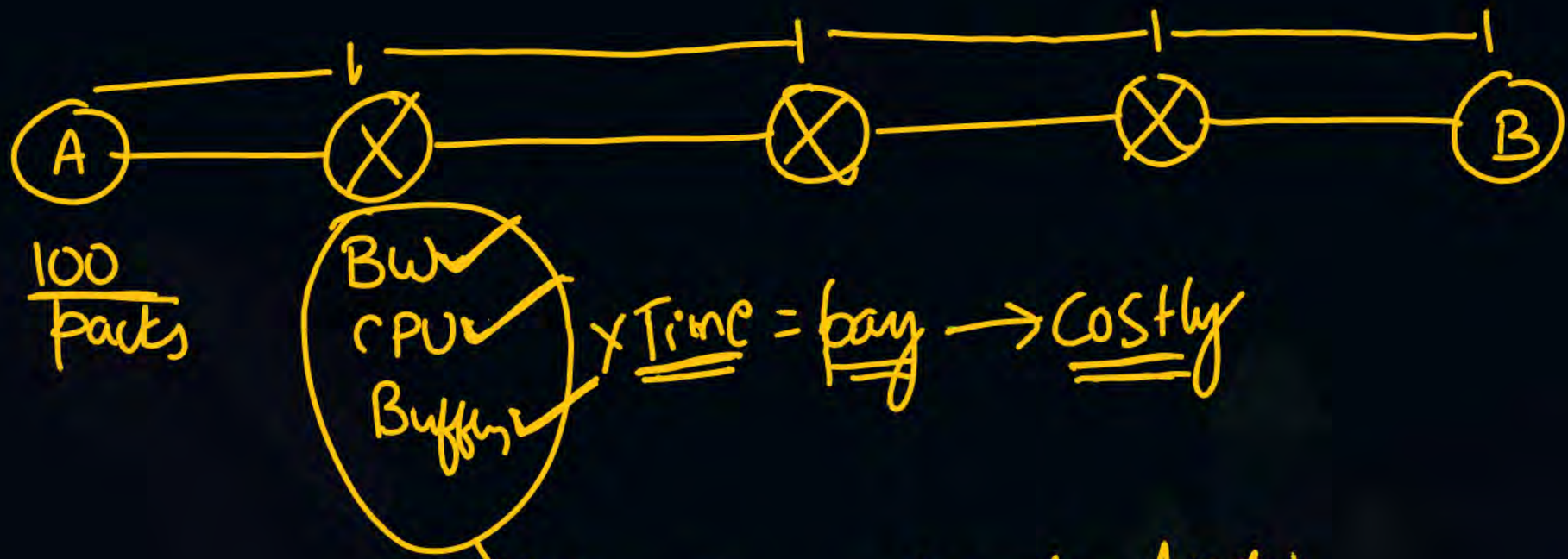
VC :-

Imagine 100 people trying to climb a mountain

First person will take a rope and climb the mountain.

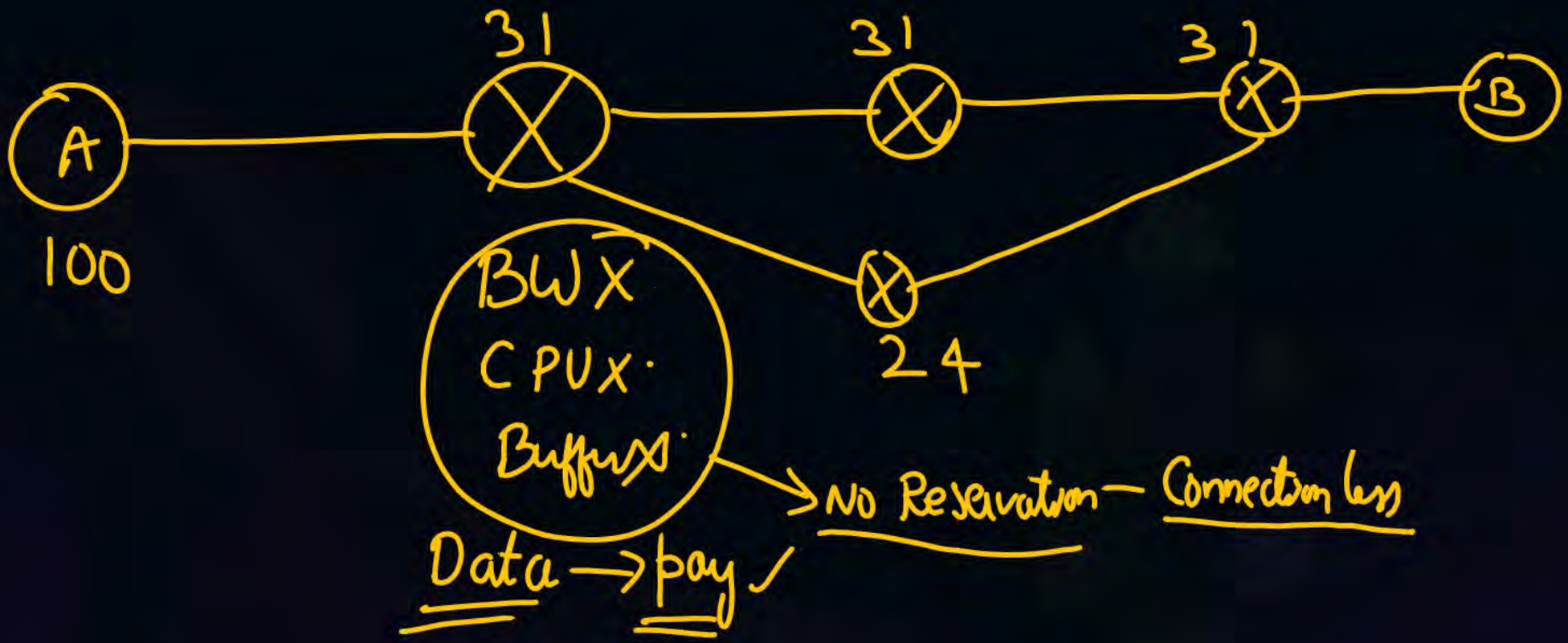
All the remaining people will use the rope to climb the mountain.

VC


A
B
100 Packets
BW
CPU
Buffer
x Time = pay → Costly
Reservation → Connection Oriented

Datagram

VOIP



Datagram:-

All the packets go independently. No reservations are made

Telephone — VC

WA call — datagram

XVOIP → Voice over internet protocol. X

VC

(i) only I packet contains global header. Remaining packets will have local headers to identify the connection

DG

1) global headers are required on all the packets, because all the packets go independtly.

VC

2) It is Connection Oriented.

Buffer / BW / CPU are reserved

DG

2) It is Connectionless.

Buffs / BW / CPU are not reserved

VC

3) Data follows the same path. All the packets appear in order

4) Highly reliable

5) Costly

6) we pay per time

Ex:- ATM ✗✓

DG

3) Data may follow diff paths, ∴ They may appear out of order

4) unreliable

5) Not Costly

6) we pay per data

Ex:- IP ✓ → uses datagram service

THANK - YOU